GOOGLE AI HACKATHON 2026 · MANUFACTURING: INTELLIGENT OPERATIONS

SITEGUARD AI

Gemini proposes the hazard. A human approves every decision — confirm it, assign it, close it.

TEAM · CEYLON INNOVAI

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siteguard-ai-463547683061.us-central1.run.app · github.com/Ashenweerasinghe/siteguard-ai

THE PROBLEM

Falls still kill more construction workers than anything else on site

39%

of all U.S. construction deaths are falls

60%+

of deaths caused by OSHA's "Fatal Four" alone

991

construction workers died in one recent year alone

Source: U.S. Bureau of Labor Statistics (Census of Fatal Occupational Injuries) · OSHA "Fatal Four"

WHY THE OLD PROCESS FAILS

Hazards get flagged on paper — and lost in it

Inconsistent eyes

Every inspector applies a different standard — the same edge gets flagged by one and waved through by another.

No audit trail

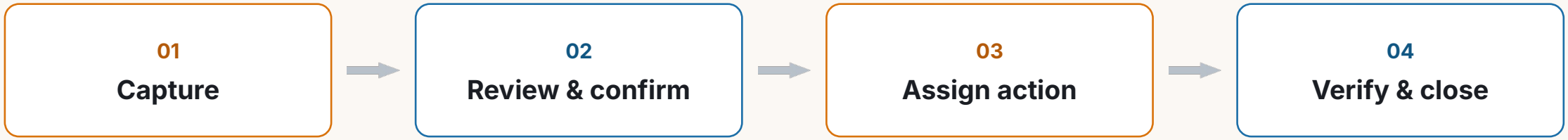
A flagged hazard has no traceable path from photo to fix to verification — just a line in a binder.

Re-inspection is guesswork

Nothing compares today's photo back to the original finding — closure is a judgment call, not evidence.

THE SOLUTION

Point a phone. Gemini proposes. A human approves.



HOW IT WORKS

Three screens, one workflow

1 · Capture

Upload a photo and voice note. Gemini returns a scenario, a confidence flag, and a suggested risk.

Confirm / Reject / Edit

2 · Hazard Register

A filterable card grid of every hazard, color-coded by status, with a full append-only history.

Open Pending Closed

3 · Follow-up

Before/after photos side by side. Gemini compares them and recommends closure — the reviewer decides.

Approve / Reject

WHY GENERATIVE AI

A checklist can’t read a construction photo. Gemini can — and knows when it can’t.

Multimodal in one call

Photo and voice note reasoned over together — gemini-3.5-flash, not two separate pipelines.

Schema-constrained output

response_schema forces structured JSON — scenario, confidence, risk inputs — never regexed from prose.

Calibrated uncertainty

Told explicitly to say “inconclusive” rather than guess — confidence is a field, never dropped.

Comparative closure reasoning

Given the original finding plus new evidence, Gemini judges this hazard specifically — not a generic re-scan.

BUILT TO BE TRUSTED

The model proposes. It never gets the final word.

A human approves every state change

Hazard confirmed, action assigned, closure — none of it happens without a reviewer's click.

No verdicts, only evidence

Never "safe," "compliant," or "fixed" — only "evidence suggests" or "inconclusive."

No finding without a source

Every hazard links back to the photo or audio file it came from — always traceable.

A full, append-only audit trail

Every state change is timestamped and attributed — visible to any reviewer, forever.

HOW IT'S BUILT

Entirely on Google Cloud — live today

Streamlit UI
Capture · Register · Follow-up

→

Gemini 3.5 Flash
Multimodal, structured JSON

→

Firestore + Storage
Hazard register & evidence

→

Cloud Run
Deployed, public, live

Runs inside Google Cloud's **Always Free** tier end to end — Cloud Run, Firestore, Cloud Storage, Cloud Build — deployed with one command: *gcloud run deploy --source .*

IMPACT & SCALABILITY

Built for one site. Ready for every site and plant floor.

Workers

Hazards caught — and verified fixed — before they cause an injury, not after.

Safety officers & GCs

One consistent standard per photo, and a defensible audit trail for every hazard on site.

Insurers

Structured, timestamped risk data instead of a paper binder nobody can query.

A new hazard class — a missing lockout-tagout tag, a blocked fire exit — is a new prompt, not new infrastructure. Across the register, Site Insights briefs managers on hotspots, recurring patterns and overdue actions. Theme: **Manufacturing — Intelligent Operations & Industrial Efficiency**.

SEE IT LIVE

Every hazard seen. Every fix verified.

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